

Intent-Driven Task Routing and Cooperative Decision Making in Distributed Agent Networks

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Abstract

This paper introduces a novel framework for Intent-Driven Task Routing and Cooperative Decision Making in Distributed Agent Networks. The framework enables a network of autonomous agents to dynamically route tasks and make cooperative decisions based on shared intent, leading to improved efficiency and adaptability in complex environments. The proposed architecture leverages a distributed ledger for intent propagation and a consensus mechanism for decision-making, ensuring transparency and trust among agents. The framework is evaluated through extensive simulations and real-world deployments, demonstrating its effectiveness in handling dynamic tasks and adapting to changing network conditions. The results show that the proposed framework significantly outperforms traditional task routing and decision-making methods in terms of task completion time, resource utilization, and system resilience.